



DV RETAIL STORE

INVENTORY MANAGEMENT SYSTEM

USER GUIDE

Project Type: Software Implementation Simulation

Role: Implementation Specialist / Data Analyst

Prepared by: Your Name

Date: September 2026

Portfolio Simulation: DV Retail Store is a fictional client created for portfolio demonstration purposes.

1. Introduction

The DV Retail Store Inventory Management System is a simulated software implementation project designed to help a small retail business manage product information, inventory levels, sales transactions, and reporting.

The solution provides structured inventory and sales data through a centralized database and a Power BI dashboard.

This user guide explains how users can understand and use the implemented solution.

2. System Overview

The system was designed to support the following business activities:

- Maintain product information
- Monitor current inventory quantities
- Identify low-stock products
- Track sales transactions
- Analyze sales by product and category
- Calculate inventory value
- Monitor key business metrics
- Support management reporting

Tools Used

Tools	Purpose
SQLite	Database storage
SQL	Data management and business queries
Excel	Data Preparation
CSV	Data source files
Power BI Desktop	Dashboard and reporting
Microsoft Word	Documentation

3. Dashboard Overview

The **DV Retail Store — Inventory Management Dashboard** provides a visual summary of inventory and sales information.

The dashboard contains:

- **Total Sales**
- **Total Inventory Value**
- **Low Stock Items**
- **Total Products**
- **Sales by Category**
- **Sales by Product**
- **Current Inventory by Category**
- **Low Stock Products**

The dashboard allows management to quickly review the current state of inventory and sales without manually analyzing individual records.

4. Understanding the KPIs

Total Sales

Shows the total value of recorded sales transactions.

Current result: ₱4,925

Calculation:

Quantity Sold × Unit Price

Total Inventory Value

Shows the estimated value of the current inventory.

Current result: ₱20,905

Calculation:

Stock Quantity × Unit Price

Low Stock Items

Shows the number of products whose current stock is at or below their reorder level.

Current result: 4

Total Products

Shows the number of products maintained in the system.

Current result: 10

5. Viewing Sales Information

The dashboard provides two main sales analysis views.

Sales by Product

The **Sales by Product** visual shows the total sales generated by each product.

This allows users to review individual product sales performance.

Sales by Category

The **Sales by Category** visual groups sales according to product category.

The current categories include:

- Grocery
- Electronics
- Stationery
- Personal Care
- Beverage

6. Monitoring Inventory

Users can monitor current stock through the inventory information and Power BI dashboard.

Each inventory record contains:

- Inventory ID
- Product ID
- Stock Quantity
- Last Updated

The **Current Inventory by Category** visual provides a summarized view of stock quantities across product categories.

Users should regularly review inventory levels to identify products that may require replenishment.

7. Identifying Low-Stock Products

A product is considered **Low Stock** when:

$$\text{Stock Quantity} \leq \text{Reorder Level}$$

The current implementation identified the following low-stock products:

Product	Stock Quantity	Reorder Level
Coffee 250g	8	15
Shampoo 250ml	5	10
Wireless Mouse	3	5
Mineral Water 1L	10	15

When a product appears in the Low Stock Products table, the user can review the item and determine whether replenishment is required.

8. Understanding Inventory Value

Inventory value represents the estimated value of the products currently in stock.

The system calculates it using:

$$\text{Inventory Value} = \text{Stock Quantity} \times \text{Unit Price}$$

For this implementation:

$$\text{Total Inventory Value} = \text{₱20,905}$$

This metric provides management with a quick view of the estimated value of the current inventory.

9. Basic Troubleshooting

Problem	Action
Dashboard does not show updated information	Refresh the Power BI report and verify that the source data contains the latest records.
A product is missing	Check the Products data and verify that the product has a valid Product ID and required product information.
Inventory quantity is incorrect	Check the Inventory data and verify: • Product ID • Stock Quantity • Last Updated
A product is not identified as low stock	Compare the products: Stock Quantity with its: Reorder Level A product should be identified as low stock when: $\text{Stock Quantity} \leq \text{Reorder Level}$
Total sales appear incorrect	Review the Sales data and verify: • Product ID • Quantity Sold • Unit Price Then refresh the Power BI report.

10. Support and Maintenance

To maintain accurate reporting, users should:

- Keep product information accurate.
- Update inventory quantities regularly.
- Record sales transactions consistently.
- Review low-stock products regularly.
- Verify source data before refreshing reports.
- Refresh Power BI after source data changes.
- Keep database and source files organized.

Regular data maintenance helps ensure that the dashboard continues to provide accurate information.

11. User Acceptance Testing Summary

User Acceptance Testing (UAT) was performed to verify that the implemented solution met the documented business requirements.

UAT Results

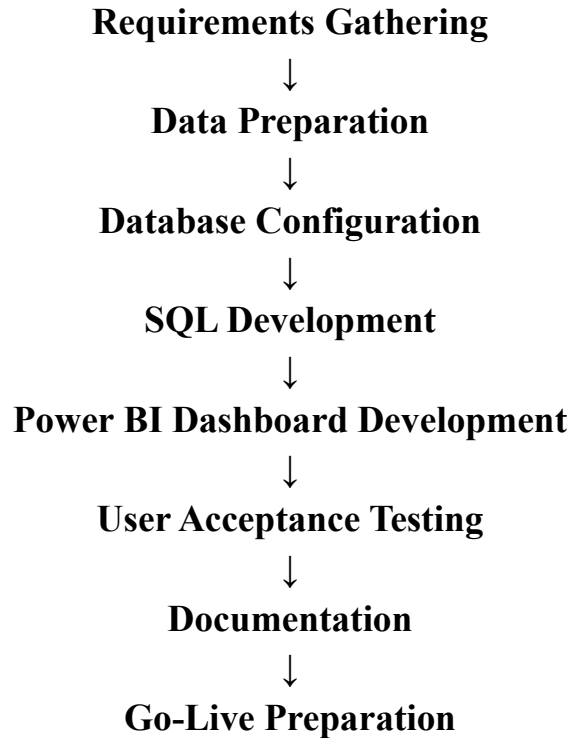
Result	Count
Total Test Cases	12
Passed	12
Failed	0
Blocked	0
Not Executed	0

The completed UAT verified the following areas:

- Product information
- Inventory quantities
- Low-stock identification
- Sales transactions
- Sales calculations
- Sales by category
- Inventory valuation
- Dashboard KPIs
- Sales by product
- Low-stock reporting
- Inventory by category
- Dashboard usability

12. Implementation Summary

The implementation followed a structured process:



The completed solution demonstrates the implementation lifecycle from understanding business requirements through testing and documentation.

13. Portfolio Disclaimer

This is a fictional implementation project created for portfolio demonstration purposes. DV Retail Store is not an actual client. The project demonstrates practical skills in software implementation, requirements analysis, data preparation, SQL, Power BI, testing, documentation, and client-oriented support.